CCP6224 Object Oriented Analysis and Design
Lab 9 – UML Structural Diagrams

Lab outcomes

By the end of today's lab, you should be able to

- differentiate and draw different UML structural diagrams
- interpret scenarios and statements and translate them to proper UML diagrams
- identify different relationships between objects and represent them graphically with a UML class diagram

UML Structural Diagrams

1. Identify the use-cases and actors in the software description below and draw the resulting UML use-case diagram

> You are developing a car rental software system for a store. The details of the software portion are as follows:
>
> - Rents only cars. The store does not sell anything
> - There are various car models and manufacturing year of cars available.
> - The rental charge varies by the models and the manufacturing year.
> - Each customer has a separate membership and must login to perform transactions (e.g. search, select, book, reservc etc)
> - The store takes RM1000 as registration deposit
> - The deposit is refunded when the car is returned.
> - Employees handle rental transactions made by customers.
> - Employees also handle the updating of information on available cars
> - Employees and managers monitor sales performance by viewing daily and monthly logs respectively
> - Customers have option to purchase add-on insurance from a insurance company.

2. Draw a UML class diagram to represent the following statements:

"The orchestra is made up of several sections. Each section can be broken down into two groups called chairs. Each chair consists of several musicians (players)"

You can ignore methods and attributes in the diagram and focus solely on the object class.

3. Draw a class diagram representing the relationship between parents and children. Take into account that a person can have both a parent and a child AND also BE a parent and a child. Annotate associations with roles and multiplicities

4. What relationship is appropriate between the following classes: aggregation, inheritance, or neither?
   a) Student-Freshman
   b) Student-Professor
   c) Car-Door
   d) Truck-Vehicle
   e) TrafficSign-Color

5. Consider a system that includes a Web server and two database servers. Both database servers are identical - The first acts as a main server, while the second acts as a redundant backup in case the first one fails. Users use Web browsers to access data through the Web server. They also have the option of using a proprietary client that accesses the databases directly. Draw a UML *deployment diagram* representing the hardware/software mapping of this system.

6. Draw a class diagram to represent the statement "*A car with an engine can have many wheels. A wheel has properties like size, tread width, rimsize and type*"; Then draw an object instance diagram to show how a Hyundai Elantra (of type *car*) has 5 wheels.

7. Consider an airline reservation system with classes Passenger, Itinerary, Flight, and Seat. Consider a scenario in which a passenger adds a flight to an itinerary and selects a seat. What responsibilities and collaborators will you record on the CRC cards as a result?

8. You are given the following scenario for a library loan system
   - The system keep tracks of when items (books, magazines, journals, references) are borrowed and returned
   - The system must support librarian work
   - The library is open to staff and active students
   - Staff may borrow up to 20 different items for up to a month
   - Students may borrow up to 10 different items for a week

   Based on the scenario above, draw a simple class diagram for the system. What can/will be recorded on CRC cards if they are used?